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ORIGINAL ARTICLE

Concurrent and Prospective Screening for Problem Drinking Among College Students

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Purpose: **Previous studies suggest that combining the CAGE questionnaire with the Perceived Benefit of Drinking Scale (PBDS), information about an adolescent's use of tobacco, and best friend's drinking pattern is a useful composite screening measure for problem drinking. The present study was undertaken to evaluate this composite screening measure prospectively as a predictor of subsequent problem drinking among late adolescents across 3 years of college.**

Methods: **A random sample of 452 college freshmen entered a longitudinal study of alcohol use at the beginning of their freshman year. A total of 184 (58%) completed follow-up measures of alcohol use 32 months later. Outcome measures included the quantity and frequency of alcohol use and a composite measure of specific alcohol-related problems.**

Results: **CAGE scores, PBDS scores, tobacco use, and best friend's drinking patterns as reported at college entry together explained 33% of the variance in the quantity/frequency measure and 37% of the variance in the alcohol-related problems measure from the end of the junior year. These same variables as reported at the end of the junior year explained 50% of the variance in the quantity/frequency measure and 61% of the variance in the alcohol-related problems measure. The composite screening measure as reported at college entry had a sensitivity of 73%, specificity of 70%, positive predictive value (PPV) of 63%, and negative predictive value (NPV) of 78% for students at high risk for problem drinking at the end of the junior year. A similar concurrent composite screening measure consisting of the same variables reported at the end of the junior year had a sensitivity of 88%, specificity of 56%, PPV of 60%, and NPV of 83% for high-risk drinkers. A total of 70–73% of students could be correctly categorized by each composite screening measure. These composite screening tests had significantly better test characteristics than the CAGE or PBDS alone.**

Conclusions: **College students' responses to the CAGE, PBDS, tobacco use, and their friends' drinking remain consistent over 3 years and correlate with concurrent and future risk for problem drinking. These variables explain significant variance in drinking and alcohol-related problems and may constitute a useful screening measure for current and future problem drinking.**

KEY WORDS:
Problem drinking
Health problems
Screening
Adolescents
College students

Identification of problem drinking among late adolescents and young adults is complicated, because heavy drinking is endemic in these age groups (1,2). Alcohol consumption by college students has remained remarkably consistent over the past 20 years (1–8). Frequent and heavy drinkers place themselves and others at increased risk for a variety of adverse consequences. Problems owing to drinking that are frequently experienced by college students include: personal injury, accidents, missing class, blackouts, legal difficulties, poor academic performance, acquaintance rape, sexually transmitted diseases, and unplanned pregnancy (1,3,5,9–11). A recent United States Department of Education student survey at 96 colleges found that 36% of all students drove while intoxicated on at least one occasion in the past year, 33% got into an argument or fight, 30% missed class,

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and 23% performed poorly on a test because of drinking (12). In another study, almost 30% of college students reported loss of hours of normal functioning while recovering from drinking during the preceding week (3).

Although it is easiest to identify substance abuse in those patients who are most severely affected, the more difficult challenge before health care providers is to identify individuals early in their involvement and to intervene in a timely manner. There are likely to be many late adolescents and young adults who suffer an adverse health or psychosocial outcome as a result of drinking but who do not meet traditional criteria for alcoholism. The Guidelines for Adolescent Preventive Services (GAPS) established by the American Medical Association recommends both primary and secondary preventive strategies to reduce adolescents' use of alcohol, tobacco, and other drugs, including screening of all adolescents and the use of brief counseling and referral, as needed (13).

The technology needed to screen for alcohol use problems among adolescents and young adults in the primary care setting remains underdeveloped. Reliance on behavioral dysfunction and psychosocial consequences of alcohol use as indicators of problem drinking may cause the physician to miss the majority of late adolescents and young adults who are occasional users, those in the early phases of alcohol use, or those who are at risk for future problem drinking. Health care providers may not be able to devote the time needed for a detailed psychosocial and behavioral history that would identify at-risk patients. Several brief clinical screening measures have been recommended as suitable for the primary care setting, including the CAGE questionnaire (14,15), the Perceived Benefit of Drinking Scale (PBDS) (16,17), the Adolescent Alcohol Involvement Scale (18), the Brief Michigan Alcoholism Screening Test (19), and the Alcohol Use Disorders Identification Test (AUDIT) (20). Klitzner et al. suggested that clinical questionnaires must meet several criteria to be useful to the practicing pediatrician: 1) the instrument must have an acceptable level of discriminant validity, 2) it must be simple and nonthreatening to maximize compliance and must be short enough to be completed in a brief period of time, and 3) it must be easily interpreted to facilitate integration into routine office practice (21).

The CAGE questionnaire, a screening instrument commonly used in health care settings (14,15), has been found to be significantly correlated with reported alcohol consumption among college students (22). However, the CAGE may not be an adequate screening instrument for problem drinking among college students when used alone (23,24). An alternative approach to screening for alcohol abuse uses the adolescent's perceptions of why he or she drinks. The PBDS probes the adolescent's view regarding the specific reinforcements received from drinking (16,17). Although the PBDS has not been used extensively with college students, it has been shown to correlate with frequent binge drinking among college freshmen (25,26). Preliminary studies suggest that the CAGE may be more useful with older male adolescents and that the PBDS may be more useful with older female adolescents (26,27).

A previous study conducted with college students at the beginning of their freshman year reported that combining the CAGE with the PBDS, the adolescent's use of tobacco, age of first drinking, and the drinking pattern of the student's best friends explained significant variance in current drinking patterns and the occurrence of alcohol-related problems (26). Together, these items established a composite screening measure for problem drinking with significantly better test characteristics than either the CAGE or PBDS alone. Further study suggested that this composite screening measure prospectively explained significant variance in alcohol use and alcohol-related problems among late adolescent women during their 1st year of college (27). Additional research is needed to identify brief clinical screening measures which are useful for prospectively identifying adolescents who go on to develop problem drinking. The present study was undertaken to prospectively evaluate a composite clinical screening measure as a predictor of alcohol-related problems and problem drinking among male and female late adolescents across 3 years of college. This study builds on two previous studies using the same sample of college students at 3-year follow-up (26,27).

Methods

Sample and Procedure

A random sample of 452 freshmen students was administered a confidential questionnaire as part of orientation programs conducted in each dormitory during the first 3 weeks of the fall semester. Participants were 16–20 years of age [mean 17.9 years, standard deviation (SD) = 0.5] A total of 91% of subjects were non-Hispanic white, 3% African-American, 2% Hispanic, and 4% Asian. Study participants did not differ significantly from the remainder

of the freshman class regarding age and race. All students voluntarily participated in this study, which was approved by the Institutional Review Board, following explanation of their rights as participants and the procedures for participation by one of the authors (MJW). All study materials and participant responses remained in the possession and confidence of the same author. Confidential follow-up questionnaires were distributed by campus mail to each participant during early April of their junior year (32 months later). Of the original 452 participants, 318 (70%) remained enrolled at the same university and were eligible for the follow-up study. Participants were asked to complete the questionnaires at a private time of their choosing, place them in a sealed unmarked envelope, and anonymously return them by campus mail to the first author's office. Approximately 3 weeks after the first mailing, a reminder letter and second copy of the questionnaire were mailed to those not yet responding. Those completing the follow-up questionnaire received a $10 coupon for use at the campus bookstore. Complete follow-up questionnaires were received from 184 participants (58%) giving an overall participation rate of 41%.

Measures

Adolescent problem drinking can take a variety of forms not limited to traditional conceptualizations of adult alcoholism (29). Frequent or heavy drinking, drunkenness, and engaging in high-risk behaviors while drinking can all be considered forms of adolescent problem drinking. The specific drinking variables assessed in this study were modeled after those used in other studies on adolescent alcohol use (1,8,23,24,29–31). The first was a quantity/frequency measure computed from responses to questions regarding typical quantity and frequency of drinking. Students were asked to report on how many days in the past month they had drunk alcohol, and, on average, how many drinks they had on each occasion of drinking. A drink was defined as equal to one beer, one wine cooler or glass of wine, or one ounce of liquor in a mixed drink. Students' responses to questions regarding frequency and average quantity of drinking per occasion in the past month were multiplied to create a quantity/frequency measure.

Second, a composite measure was created that reflected the various ways in which adolescents might experience problems related to alcohol use. The specific problems were chosen from other studies of college students and represent the spectrum of the most common outcomes associated with drinking in this age group (1,8,12,23,24). These problems are directly related to drinking alcohol and may contribute to significant risk for early medical or psychosocial consequences. Students were asked to indicate whether the following had happened to them in the past year: "Has your drinking ever caused you to have a hangover? Have you ever awakened the morning after some drinking and found that you could not remember part of the evening? Has your drinking ever caused you to get behind in schoolwork? Has your drinking ever caused you to have arguments with friends? Has your drinking ever caused you to engage in unplanned sexual intercourse? Has your drinking ever caused you to damage property? Have you suffered a physical injury because of drinking? Has your drinking ever caused you to get into physical fights? Have you had any contact with the police or legal troubles as a result of drinking? Have you had a medical illness because of drinking?" For each student, positive responses to these questions were summed to create an alcohol-related problems measure ranging from 0–10.

The third indicator of problem drinking related to frequency of drunkenness. On a 5-point response scale ranging from *never* to *always*, students were asked how often they drink to get drunk.

The remainder of the questionnaire included the CAGE questions, the PBDS, and self-report of tobacco use and peer group drinking. Students were asked to describe the drinking habits of their best friend as either: a) nondrinker, b) light, c) moderate, or d) heavy social or problem drinker. Tobacco use was assessed with questions about both cigarette and chewing tobacco use. Students were considered current tobacco users if they reported any use of tobacco products in the past month.

The CAGE questions have proven useful in screening for alcoholism (14,15). The questions focus on *cutting down* on drinking, *annoyance* at criticism by others about drinking, *guilty* feelings about drinking or something done while drinking, and the use of an *eye-opener*. The questions may be self-administered. Although a positive response to the CAGE questions is not diagnostic of alcoholism, a score of 2 or greater is highly suspicious and warrants further evaluation (14,15).

The PBDS is a brief clinical screening instrument useful in identifying adolescents who may be alcohol abusers. It consists of endorsing as true or false each of the following five statements: "Drinking helps me forget; drinking helps me be friendly; drinking helps

me feel good about myself; drinking helps me relax; and drinking helps me be friends with others who drink." Reliability and validity of the PBDS with younger adolescents are well established (16,17). It assesses the benefits an adolescent attributes to drinking, and is not dependent upon direct information about drinking patterns or negative consequences resulting from drinking. A full evaluation of potential drinking problems is strongly recommended for those scoring 3 or greater (16,17).

Statistical Analysis

Chi-square, and Student's t tests were used to test for differences between respondents and nonrespondents to the follow-up questionnaire. Pearson correlation coefficients were calculated for the quantity/frequency and alcohol-related problems measures at the end of the junior year compared to entry into college. Correlation coefficients were also calculated between CAGE scores, PBDS scores, tobacco use, and the student's best friends' drinking patterns as reported at both times and the quantity/frequency and alcohol-related problems measures at the end of the junior year.

Using a procedure similar to that in other studies of college student and adolescent problem drinking (1,8,23,24,29–31), students were divided into three groups: 1) current nondrinkers, 2) low-risk drinkers, and 3) high-risk drinkers based on their responses to the follow-up questionnaire at the end of their junior year. Nondrinkers were those who reported no alcohol use in the past month. Students were classified as at high risk for problem drinking if they: a) were in the top 25% on the quantity/frequency measure, or b) sustained four or more alcohol-related problems during their junior year, or c) reported drinking to get drunk more than half the time. These criteria were selected on empiric grounds to model other studies which had attempted to define problem drinking among college students on the basis of self-reported behaviors (1,8,23,24,29–31). Frequent or heavy drinking, drunkenness, and engaging in high-risk behaviors while drinking can all be considered forms of adolescent problem drinking. The top quartile of students in this study were selected as an empiric quantity/frequency cutoff as: 1) they represented those who reported drinking an average of 48 or more drinks per month (or 10+ drinks/week), a level of drinking used in other studies to define problem drinking (23,24), and 2) this percentile approximates the portion of students reported as frequent heavy drinkers in other studies on college drinking (1,8,23,24,31).

Standard multiple regression analysis with simultaneous entry of predictor variables was used to test the hypothesis that the components of the composite screening measure would both explain significant variance in subsequent drinking and the alcohol-related problems measure. Simultaneous entry of the predictor variables was preferred over a stepwise approach, as the purpose of this study was to evaluate the overall composite screening measure and not to test the individual predictors. There were concerns that identification of a highly specific predictor variable would have less generalizability to other college populations and that an advantage of the multifactorial composite measure being evaluated may be greater applicability to differing college populations.

Results

Descriptive Data

Respondents (n = 184) to the follow-up questionnaire and nonrespondents (n = 134) did not differ significantly with respect to age, race, or gender. Nonrespondents had significantly higher scores on both the quantity/frequency of drinking measure ($t = -4.9$, $df = 209$, $p < .001$) and the alcohol-related problems measure ($t = 3.4$, $df = 239$, $p < .001$) than respondents at entry into college. Those students no longer enrolled at the university (n = 134) were not significantly different from those remaining through their junior year on either the quantity/frequency of drinking or alcohol-related problems measures.

The quantity/frequency measure at follow-up had a range of 0–225 (mean 31, SD = 38, median 16). Male students had significantly higher follow-up quantity/frequency scores than did females (mean scores 41 vs. 22, $t = -3.3$, $p < .001$). It is noteworthy that the mean drinking index score of 41 for males averages 1.4 drinks/day. Drinking quantity/frequency in the junior year was significantly correlated with quantity/frequency at entry into college (r = .69, $p < .01$). In their junior year, 25% of the subjects had drunk five or more drinks on four or more occasions in the past month, and 23% reported drinking to get drunk more than half the time or always. A total of 11% of the subjects were nondrinkers (8% of males and 13% of females) at the end of their junior year.

A substantial portion of the juniors reported that they had sustained many alcohol-related problems

Table 1. College Juniors Reporting Alcohol-Related Problems During the Past Year (%)

	Total (n = 184)	Male (n = 86)	Female (n = 98)
Hangover	63	63	62
Blackouts	41	41	41
Got behind in schoolwork from drinking	22	24	20
Argue with friends because of drinking	21	28	15
Unplanned sex while drinking	16	17	14
Damage property because of drinking	10	17	3
Alcohol-related injury	8	9	6
Physical fighting because of drinking	7	13	1
Alcohol-related legal problems	3	5	1
Alcohol-related illness	1	1	1

during the past year (Table 1). As many as 35% of subjects reported driving under the influence at least once in the past year. The alcohol-related problems measure had a range of 0–10 (mean 2.7, SD = 2.6, median 2). Male students sustained significantly more alcohol-related problems during their junior year than did female students (mean scores 3.2 vs. 2.3, $t = 2.3$, $p < .03$). A total of 22% reported four or more alcohol-related problems during their junior year. The number of alcohol-related problems reported during the junior year was significantly correlated with the number sustained prior to entering college ($r = .61$, $p < .01$).

Prediction of Problem Drinking

There were 20 (11%) nondrinkers, 93 (51%) low-risk drinkers, and 71 (38%) high-risk drinkers. Of the 71 high-risk drinkers, 47 (66%) met two or more of the above criteria for high-risk drinking. Of the 24 high-risk drinkers who met only one criterion, 10 were in the top quartile on the quantity/frequency measure, 9 sustained four or more alcohol-related problems, and 5 reported drinking to get drunk more than half the time. The three criteria for high-risk drinking were all significantly correlated with each other as follows: quantity/frequency criterion with alcohol-related problems criterion ($r = 0.77$, $p < .01$), quantity/frequency criterion with drinking to get drunk criterion ($r = 0.58$, $p < .01$), and alcohol-related problems criterion with drinking to get drunk criterion ($r = 0.75$, $p < .01$).

Both the quantity/frequency and alcohol-related problems measures from the end of the junior year were significantly correlated with CAGE scores, PBDS scores, tobacco use, and the student's best friends' drinking patterns as reported at both entry into college and at the end of the junior year ($ps < .001$) (Table 2). Chi-square analyses of the individual questions within the CAGE and PBDS found that significantly more high-risk than low-risk drinkers responded with a positive answer to essentially every question as reported at both Time 1 and Time 2 (Table 3). Significantly more high-risk than low-risk drinkers had CAGE scores of 2 or greater and PBDS scores of 3 or greater at both entry into college and 3 years later. Similar results were found for both male and female students.

Using standard multiple regression analysis, the set of variables including CAGE scores, PBDS scores, tobacco use, and the student's best friends' drinking pattern as reported at college entry together explained 33% of the variance in the quantity/frequency measure and 37% of the variance in the alcohol-related problems measure from the end of the junior year (Table 4). These same variables as reported at the end of the junior year explained 50% of the variance in the quantity/frequency measure

Table 2. Pearson Correlation Coefficients Between Quantity/Frequency and Alcohol-Related Problems Measures as Reported at the End of the Junior Year and Independent Variables (n = 184)

	Quantity/ Frequency Measure		Alcohol-Related Problems Measure	
	r	p	r	p
Beginning of college				
CAGE scores	.34	.0001	.29	.0001
PBDS scores	.30	.0001	.44	.0001
Friends' drinking pattern	.40	.0001	.49	.0001
Tobacco use	.50	.0001	.37	.001
End of junior year				
CAGE scores	.47	.0001	.56	.0001
PBDS scores	.27	.001	.48	.001
Friends' drinking pattern	.55	.0001	.57	.0001
Tobacco use	.53	.0001	.50	.0001

Table 3. Chi-Square Analyses of CAGE and PBDS Questions with High- and Low-Risk Drinkers at Follow-up

	Entry into College			End of Junior Year		
	Low-Risk Drinker (n = 103) (%)	High-Risk Drinker (n = 71) (%)	p	Low-Risk Drinker (n = 103) (%)	High-Risk Drinker (n = 71) (%)	p
CAGE questions						
Cutting down	7	22	.004	12	41	.0001
Feeling annoyed	7	11	.35	5	15	.03
Feeling guilty	8	25	.003	14	25	.05
Using eye-opener	5	16	.02	4	30	.0001
CAGE score ≥2	5	25	.003	8	29	.0001
PBDS questions						
Drinking to forget	8	23	.005	8	28	.0003
Drinking to be friendly	25	56	.0001	28	60	.0001
Drinking to feel good	6	18	.02	6	19	.006
Drinking to relax	28	62	.0001	55	80	.0005
Drinking with others	13	25	.05	15	23	.15
PBDS score ≥3	14	29	.0001	13	33	.0001

and 61% of the variance in the alcohol-related problems measure (Table 5).

Separate standard multiple regression analyses were performed for male and female students using the same independent variables from both Time 1 and Time 2 as predictors of variance in the dependent measures at the end of the junior year (Tables 4 and 5). The regression models explained more variance among male students than female students. As predictors of future alcohol use or related problems, CAGE scores and tobacco use were significant factors in the regression models only for male students (Table 4). Conversely, PBDS scores explained significant variance only for female students.

Table 4. Multiple Regression Analysis Predicting Scores on the Quantity/Frequency and Alcohol-Related Problems Measures at the End of the Junior Year: Independent Variables from College Entry

	Quantity/ Frequency Measure		Alcohol-Related Problems	
	β	p	β	p
All students*				
CAGE scores	.147	.04	.090	.20
PBDS scores	.107	.15	.261	.0003
Tobacco use	.384	.0001	.249	.0003
Friends drinking	.180	.02	.273	.0004
Male students†				
CAGE scores	.219	.04	.127	.20
PBDS scores	.052	.60	.186	.07
Tobacco use	.457	.0001	.300	.006
Friends drinking	.099	.40	.300	.009
Female students‡				
CAGE scores	.104	.30	.056	.60
PBDS scores	.274	.02	.389	.0004
Tobacco use	.115	.24	.038	.07
Friends drinking	.231	.04	.187	.07

* All students: quantity/frequency measure: F = 15.80; R^2 = .33; adjusted R^2 = .31; p < .0001. Alcohol-related problems measure: F = 18.79; R^2 = .37; adjusted R^2 = .35; p < .0001.

† Male students: quantity/frequency measure: F = 9.40; R^2 = .41; adjusted R^2 = .36; p < .0001. Alcohol-related problems measure: F = 10.60; R^2 = .44; adjusted R^2 = .39; p < .0001.

‡ Female students: quantity/frequency measure: F = 4.82; R^2 = .23; adjusted R^2 = .18; p < .0006. Alcohol-related problems measure: F = 8.39; R^2 = .34; adjusted R^2 = .30; p < .0001.

As predictors of current alcohol use and related problems at the end of the junior year, both the CAGE and tobacco use explain significant variance in alcohol-related problems for both genders (Table 5). The PBDS also enters the regression model as a significant predictor for alcohol-related problems among males as well as females.

Two composite screening measures were computed for each subject based on responses at a) college entry and b) the end of the junior year. This measure was computed by adding 1 point for each positive response to the CAGE items, PBDS items, using tobacco in the past month, and describing their best friends as heavy drinkers. Each item was equally weighted, as there was no compelling rationale for a weighted approach. The composite screens were compared to both the CAGE and PBDS as predictors of students at high risk for problem drinking.

Use of the composite screen significantly increased sensitivity beyond that obtained with the CAGE or PBDS alone for concurrent and prospective identification of high-risk students (Tables 6 and 7). The composite screening measure consisting of responses at college entry had a sensitivity of 73%, specificity of 70%, positive predictive value of 63%, and negative predictive value of 78% for students at high risk for problem drinking at the end of the

Table 5. Multiple Regression Analysis Predicting Scores on the Quantity/Frequency and Alcohol-Related Problems Measures from the End of the Junior Year: Independent Variables from Junior Year

	Quantity/ Frequency Measures		Alcohol-Related Problems	
	β	*p*	β	*p*
All students*				
CAGE scores	.267	.0001	.280	.0001
PBDS scores	.101	.90	.228	.0001
Tobacco use	.348	.0001	.305	.0001
Friends drinking	.365	.0001	.344	.0001
Male students†				
CAGE scores	.357	.0006	.321	.0003
PBDS scores	.007	.90	.202	.009
Tobacco use	.318	.0006	.340	.0001
Friends drinking	.295	.0009	.290	.0002
Female students‡				
CAGE scores	.137	.15	.253	.002
PBDS scores	.031	.80	.254	.004
Tobacco use	.224	.01	.211	.006
Friends drinking	.494	.0001	.412	.0001

* All students: quantity/frequency measure: $F = 43.36$; $R^2 = .50$; adjusted $R^2 = .49$; $p < .0001$. Alcohol-related problems measure: $F = 67.14$; $R^2 = .61$; adjusted $R^2 = .60$; $p < .0001$.

† Male students: quantity/frequency measure: $F = 22.72$; $R^2 = .55$; adjusted $R^2 = .52$; $p < .0001$. Alcohol-related problems measure: $F = 38.10$; $R^2 = .67$; adjusted $R^2 = .65$; $p < .0001$.

‡ Female students: quantity/frequency measure: $F = 12.90$; $R^2 = .36$; adjusted $R^2 = .34$; $p < .0001$. Alcohol-related problems measure: $F = 22.8$; $R^2 = .50$; adjusted $R^2 = .48$; $p < .0001$.

junior year (Table 6). A similar composite screen consisting of the same variables reported at the end of the junior year had a sensitivity of 88%, specificity of 56%, positive predictive value of 60%, and negative predictive value of 83% for junior students at high risk for problem drinking (Table 7). The concurrent composite screening measure had slightly better results for male students, with a sensitivity of 95% and negative predictive value of 91%. A total of 73% of junior students could be correctly categorized as being at low compared with high risk for problem drinking based on the composite screening measure at college entry, and 70% could be correctly categorized based on the composite screening measure from the junior year. These composite screening measures had significantly better test characteristics than the CAGE or PBDS alone (Tables 6 and 7).

Discussion

The development of clinically useful screening instruments for the concurrent and prospective identification of late adolescents and young adults at high risk for problem drinking is an important concern for health care providers. The approach to problem drinking used in this study seeks to identify those who use alcohol in such a manner as to place themselves at high risk for a variety of alcohol-related morbidities, regardless of their likelihood for being alcoholic. Using a definition of high risk for problem drinking which is sensitive to both quantity and frequency of use and the occurrence of alcohol-related problems, 38% of the college juniors in this study were considered to be at high risk. Defining such a large group of late adolescents as at high risk for problem drinking is supported by other studies on college student drinking (1,6,8,12,32–34). Previous studies indicate that the prevalence of problem drinking among college students ranges from 6–72%, depending on the college setting (33). Among the college students in this study, 25% reported binge

Table 6. Test Characteristics of Alcohol Questionnaires at College Entry in Screening for Junior Students at High Risk for Problem Drinking

	Score	Sensitivity (%)	Specificity (%)	Positive Predictive Value (%)	Negative Predictive Value (%)
All students					
CAGE	≥2	25	95	78	64
PBDS	≥3	55	77	64	70
Composite screen*	≥2	73	70	63	78
Female students					
CAGE	≥2	13	95	57	66
PBDS	≥3	41	83	57	72
Composite screen*	≥2	66	71	55	79
Male students					
CAGE	≥2	34	95	88	60
PBDS	≥3	20	91	67	54
Composite screen*	≥2	78	67	70	76

* Sum of positive responses to CAGE and PBDS questions, using tobacco, reporting best friends to be heavy drinkers.

Table 7. Test Characteristics of Alcohol Questionnaires at the End of the Junior Year in Screening for Junior Students at High Risk for Problem Drinking

	Score	Sensitivity (%)	Specificity (%)	Positive Predictive Value (%)	Negative Predictive Value (%)
All students					
CAGE	≥2	29	91	73	65
PBDS	≥3	63	71	62	73
Composite screen*	≥2	88	56	60	83
Female students					
CAGE	≥2	20	92	58	65
PBDS	≥3	32	90	65	70
Composite screen*	≥2	71	61	51	79
Male students					
CAGE	≥2	37	93	83	60
PBDS	≥3	37	83	70	59
Composite screen*	≥2	95	49	65	91

* Sum of positive responses to CAGE and PBDS questions, using tobacco, reporting best friends to be heavy drinkers.

drinking at least weekly and 23% reported drinking specifically to get drunk more than half the time.

This study focused upon health problems directly related to drinking alcohol and likely to be associated with significant risk for morbidity and mortality. The issues addressed take a broad view of health, and include items reflecting potential risk for medical, legal, or academic problems as a result of drinking. The occurrence of alcohol-related problems during the junior year was significantly correlated with sustaining similar problems prior to entering college. Among these students, 22% had had four or more of the health problems surveyed, and 10% had been affected by six or more. Blackouts, a symptom commonly used as an indication of alcohol abuse, were experienced by 41% of the students during their junior year.

Both the CAGE and PBDS have been used to identify current risk for problem drinking, but neither has been adequately evaluated for its use in prospectively identifying late adolescents who are problem drinkers. In this study, college juniors at high risk for problem drinking were significantly more likely to have had high-risk CAGE and PBDS scores 3 years earlier at entry into college. Results of this longitudinal study build upon previous findings which suggested that expanding on a combination of the CAGE and PBDS to include tobacco use and peer drinking patterns can be useful in prospectively identifying late adolescents at high risk for problem drinking (27). These variables as reported at college entry explained 33–37% of the variance in specific alcohol-related problems and quantity and frequency of drinking 3 years later during the junior year. The composite screening measure had significantly better test characteristics than either the CAGE or PBDS alone and was able to correctly categorize 73% of the junior students based on responses at college entry.

The present study confirms previous studies supporting the usefulness of combining the CAGE, PBDS, tobacco use, and peer drinking pattern as a screening measure for current risk for problem drinking among college students (26,27). These variables, as reported during the junior year, explained 50–61% of the variance in specific alcohol-related problems and quantity and frequency of drinking during the previous year. The composite screening measure again had significantly better test characteristics than either the CAGE or PBDS alone and was able to categorize 70% of the junior students correctly.

Previous studies have suggested that the CAGE and PBDS have limitations when used with late adolescents (23–27). Heck found that the CAGE together with the student's likelihood of choosing nonalcoholic beverages at social events, history of driving under the influence of alcohol, and having started regular use of alcohol before college were very useful in screening for problem drinking (24). Results of this study support previous findings which suggested that the CAGE may be more useful with male adolescents and the PBDS more useful with female adolescents (26,27).

It is likely that the components of the composite screening measure may be measuring different aspects of problem drinking. This is supported by the different performances of the CAGE and PBDS for female compared with male students. However, this may be of pragmatic value as this composite screen-

ing measure is intended to be used as a clinical tool and not as a psychometric instrument. Those individual items within the composite screen that have greater use with one college population may be less useful with a different population. Tobacco use and friends' drinking may be the most important predictors for the population in this study based on the regression analyses. However, it is possible that different components of this screening measure would be more critical as predictors with other college populations. Further research is needed to evaluate this type of brief multifactorial clinical screening tool with other late adolescent and young adult populations.

Brief screening questionnaires such as the CAGE and PBDS are most useful in stimulating direct discussion about alcohol use and the adolescent's self-perception of their use. These clinical aids are not intended to be diagnostic instruments; rather, they facilitate gathering information which can be used to complement the psychosocial history. Experienced interviewers will not simply ask each question within the CAGE, PBDS, or any other screening tool, but will use the areas targeted by these questions briefly to probe the critical issues behind adolescent alcohol use. For example, when adolescents acknowledge a previous attempt to cut down on drinking, this provides an excellent opportunity to explore: a) their self-perceptions of problems they themselves have noted as a result of drinking, and b) other factors which motivated their earlier attempt to change their alcohol use. When adolescents admit to feelings of guilt because of behaviors while drinking, they have a sense of the need for change and may feel motivated because of it. It is noteworthy that in this study all nine questions within the CAGE and PBDS were answered positively by a significantly greater percentage of high-risk than low-risk drinkers. Familiarity with the general content of these screening measures can help the pediatrician better understand the objectives of an alcohol use screening interview.

The results of this study support the temporal stability of commonly used indicators of adolescents at high risk for current or future problem drinking. Because the specific items within the composite screening measure correlate with drinking behaviors and the occurrence of alcohol-related problems at both entry into college and the end of the junior year it could be argued that the longitudinal relation is potentially an artifact of a concurrent relation. This situation will be found with any relatively stable predictor and is difficult to control for experimentally. It also reflects that those adolescents who are at risk for problem drinking in the future are also presently at risk.

This study relied upon students' self-report of health problems and drinking habits. Approximately 90% of the sample in this study was Caucasian from a selective southern university, limiting generalizability to other adolescent populations. Subjects who did not respond to the follow-up questionnaire had slightly higher alcohol-related problems index scores at the beginning of the freshman year than did those completing the study. This suggests that the final study sample underrepresented those students at greatest risk for the alcohol-related problems. However, the breadth of drinking habits and the wide distribution of alcohol-related problems indicates that heavy-drinking students and those sustaining numerous problems were represented in the final sample. The appraisal of alcohol-related problems was limited and did not assess the impact of each problem on the student. Empiric criteria were used to define problem drinking based on the approaches used in other studies. The difficulty in defining problem drinking compromises generalizability, particularly to college populations who display lesser or greater alcohol consumption.

College students' responses to the CAGE, the PBDS, their tobacco use, and their friends' drinking pattern remain consistent over 3 years and correlate with concurrent and future risk for problem drinking. These variables explain significant variance in quantity and frequency of drinking and in alcohol-related problems, and may constitute a useful screening measure for current and future problem drinking.

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